Lucio La Cava

PhD Student in Information and Communication Technologies | Teaching Assistant

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Education

PhD Student in Information and Communication Technologies
Rende, Italy

Dept. Computer Engineering, Modeling, Electronics, and Systems (DIMES), University of Calabria *November 2020 - Present*

- Topics: Data Science, Network Science, Machine Learning

MSc in Computer Engineering
Rende, Italy

University of Calabria *September 2017 - July 2020*

- Supervisor: Prof. Andrea Tagarelli
- Final mark: magna cum laude

BSc in Computer Engineering
Rende, Italy

University of Calabria *September 2014 - September 2017*

- Supervisor: Prof. Luigi Palopoli
- Final mark: 106/110

Experiences

Teaching assistant
Rende, Italy

Dept. Computer Engineering, Modeling, Electronics, and Systems (DIMES), University of Calabria *October 2020 - Present*

Course of Networks and Systems in Bioinformatics - Master's degree in Computer Engineering (Big Data and Bioinformatics)

Italian qualification to practice as an Engineer
Rende, Italy

Section A, Field of Information Engineering *No expiration date*

Qualifying exam passed on November 2020

Reviewer

Journals

- Social Network Analysis and Mining (SNAM)
- Information Systems (IS)
- Applied Network Science (APNS)

Conferences

- The 29th edition of the Italian Symposium on Advanced Database Systems (SEBD - Sistemi Evoluti per Basi di Dati)
  September 5, 2021 - September 9, 2021, Pizzo Calabro (Italy)

Skills

Macro-skills	Data Science, Network Science, Machine Learning, Artificial Intelligence, Cyber-Security
Programming	Python (Proficient), R (Proficient), Java (Proficient), Assembly (Familiar), C (Familiar), C++ (Familiar)
Data management	Relational DBs, NoSQL DBs, Database management systems (DBMSs), Business Intelligence (OLAP)
Languages	Italian (Native), English (Writing - C1, Reading - C1, Listening - B2, Speaking - B2)

Publications

"The Italian Civil Code Network Analysis"

La Cava L., Simeri A., Tagarelli A.

Proceedings of the First International Workshop RELATED - Relations in the Legal Domain 2021 (RELATED 2021), São Paulo, Brazil, June 25, 2021

Abstract: The regulation of private law is a focal element in the metamorphosis of society, and by gaining a broader vision of the legal domain we might grasp novel or potentially hidden nuances and characteristics. In this work, we explore the Italian Civil Code (ICC) from an unprecedented perspective based on network analysis. We develop a text processing method to identify and extract article references from the ICC and, upon these, we define network models capturing their relation structures either at a book and corpus scale. The exploitation of the main structural features of these networks leads us to unveil meaningful patterns, holding within and across the books composing the ICC. Furthermore, by leveraging a community detection task, we investigate whether the formation of a community is related to the topic coherence of its assigned articles over the portions of books involved. Our findings reveal useful indicators that may help legal experts and practitioners enhance their knowledge from a novel perspective provided by the network of article references through the ICC.

"Understanding the growth of the Fediverse through the lens of Mastodon"

La Cava L., Greco S., Tagarelli A.

Accepted for publication with Applied Network Science, Springer Open, June 12, 2021

Abstract: Open-source, Decentralized Online Social Networks (DOSNs) are emerging as alternatives to the popular yet centralized and profit-driven platforms like Facebook or Twitter. In DOSNs, users can set up their own server, or instance, while they can actually interact with users of other instances. Moreover, by adopting the same communication protocol, DOSNs become part of a massive social network, namely the Fediverse. Mastodon is the most relevant platform in the Fediverse to date, and also the one that has attracted attention from the research community. Existing studies are however limited to an analysis of a relatively outdated sample of Mastodon focusing on few aspects at a user level, while several open questions have not been answered yet, especially at the instance level. In this work, we aim at pushing forward our understanding of the Fediverse by leveraging the primary role of Mastodon therein. Our first contribution is the building of an up-to-date and highly representative dataset of Mastodon. Upon this new data, we have defined a network model over Mastodon instances and exploited it to investigate three major aspects: the structural features of the Mastodon network of instances from a macroscopic as well as a mesoscopic perspective, to unveil the distinguishing traits of the underlying federative mechanism; the backbone of the network, to discover the essential interrelations between the instances; and the growth of Mastodon, to understand how the shape of the instance network has evolved during the last few years, also when broading the scope to account for instances belonging to other platforms. Our extensive analysis of the above aspects has provided a number of findings that reveal distinguishing features of Mastodon and that can be used as a starting point for the discovery of all the DOSN Fediverse.

Talks/Conferences

"Towards Mesoscopic Structural Analysis of the Fediverse of Decentralized Social Networks"

La Cava L., Ruffo L. E., Tagarelli A.

The 9th International Conference on Complex Networks and their Applications (CNA2020), December 1-3, 2020 – Madrid